

PRAKASH RITHWIK KARRY

✉ rithwikp1612@gmail.com | ☎ +1 (765)-(701)-3197 | in www.linkedin.com/in/kprithwik/
| 📍 West Lafayette, IN

EDUCATION

Purdue University <i>Master of Science in Autonomy and Robotics</i> <i>Relevant Coursework: Autonomous Systems, Artificial Intelligence, Embedded Systems, Reinforcement Learning</i>	West Lafayette, IN, USA Aug. 2024 – (May. 2026)
National Institute of Technology Calicut <i>Bachelor of Technology in Engineering Physics</i>	Calicut, Kerala, India July. 2018 – May 2022

SKILLS & INTERESTS

Technical Skills: ROS2(Humble), NAV2, MicroROS, RTOS, CAD, 3D Printing, Git
Programming Languages: C, C++, Python, SQL, MATLAB, Simulink
Frameworks: TensorFlow, Mujoco, OpenCV, ROS, Flutter, PyTorch.

WORK EXPERIENCE

Robotics Research Intern <i>International Institute of Information Technology(IIITH)</i> - Designed and modeled a prototype Flapping Wing UAV with a modular, swappable powertrain module, using CAD modeling and 3D printing techniques. - Conducted research on error estimation, navigation, and safe deployment of an autonomous MAV in a GPS-denied indoor environment, in association with Airbus. - Developed an obstacle avoidance algorithm based on the velocity obstacle method for UTM and worked with a UWB-based RTLS in fusion with IMU for state estimation for indoor inspection.	Aug 2022 – May 2024 Hyderabad, Telangana, India
Project Engineer <i>Arka Aerospace</i> - Designed and tested the landing module for a Wearable Large UAV, with a re-configurable Chassis. - Developed robot navigation with ROS2 Humble and NAV2, using SLAM and LiDAR for mapping and path planning.	May 2022 – July 2022 Hyderabad, Telangana, India

RESEARCH EXPERIENCE

Graduate Researcher <i>Purdue University</i> - Implemented a Semi-Supervised Learning Framework (FixMatch) from scratch using PyTorch, achieving comparable accuracy benchmarks on CIFAR-10 and SVHN datasets with limited labeled data. - Developed a real-time gesture-controlled robotic arm manipulator integrating STM32 microcontroller, OpenCV, and MediaPipe with a 95% gesture recognition accuracy and motor control. - Developed deep reinforcement learning pipeline using Proximal Policy Optimization (PPO) for vision-based autonomous driving. - Developed a LiDAR-based front-view perception pipeline for vehicle detection in autonomous driving scenarios. - Developed the perception pipeline for LiDAR-vision fusion and integration with Vision-Language Model (VLM) reasoning frameworks.	Jan 2025 – Dec 2025 West Lafayette, IN, USA
Undergraduate Researcher <i>National Institute of Technology(NITC)</i> Synthesized BiFeO3 nanostructures using the Sol-Gel method and studying the effect of process parameters on the photocatalytic activities of BFO nanostructures.	Jan 2022 – May 2022 Calicut, Kerala, India

PUBLICATIONS

Design and Analysis of a Modular Flapping Wing Robot with a Swappable Powertrain Module <i>IEEE ICMA 2024</i> - Awarded the Best Paper Finalist. - The paper presents a novel approach to designing and modeling a prototype Flapping Wing UAV with a modular, swappable powertrain, it explores the efficiency and versatility of modular design in flapping wing robots.	Tianjin, China
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LEADERSHIP & ACTIVITIES

Joint Secretary at AeroUnwired Club at NITC: Organized workshops on aerial robotics for school and college students.
Head of Technical Workshop Committee Demonstrated effective leadership skills in organizing and promoting technical workshops for TATHVA, the annual technical fest at NIT Calicut.